

# CIS 613: Project / Paper Description

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## Project Information

- What *is* it?
  - Project **and** Presentation
- A handful of ideas:
  - A **paper** about software testing.
    - One specific topic,
    - Summarize & *analyze* related papers
  - A project **about** software testing.
    - Research topics
    - Specific tools
      - Creation, or
      - Exploration / Explanation
  - A project **using** software testing.
    - A project that doesn't have tests, could be
      - Personal
      - Professional
      - Open source

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## What is it?

- Project can be *anything* you want it to be that satisfies the following criteria:
  - Must be **related** to software testing,
  - Must **expand** upon and/or **synthesize** the concepts in this class,
  - Must be of sufficient **effort** (*intended to be a ½ semester effort*),
  - Must include a **written and presentation** deliverable,

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## Paper

Write a paper on a specific topic. For example:

- Testing a new type of software (e.g., AI)
- Applying a tool to a new domain

~~Summarize or analyze paper(s) from applicable conferences:~~

- ~~[International Conference on Software Testing, Verification, and Validation](#)~~
- ~~[International Workshop on Automaton of Software Test](#)~~
- ~~Papers from other applicable IEEE and/or ACM conferences.~~

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## Project: **About**

Prove MC/DC coverage is achievable for all non-redundant and non-conflicting sets of conditionals.

Create a tool to perform some kind of testing not easily done.

Explore and explain a tool to perform some kind of testing.

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## Project: **Using**

Create a tests for:

- A project from work that doesn't have tests,
- A personal project that could benefit from tests,
- An open source system that could benefit from tests.

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## Advice

- If you aren't sure about your idea: Talk to me!
- Don't do *too little*. But don't try to do *too much*!
- Find something **you** are *interested in or would benefit you*.

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## Previous Papers & Projects

- **Paper**: Challenges in the Elicitation and Determination of Precise Requirements from Animal Stakeholders (from CIS 612 class)
- **Project, About**: Program to measure cyclomatic complexity (from CIS 611 class)
- **Project, Using**: Applying methods from class to test software projects

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## In-Class Activity

- Group up with ~2-3 people around you
- Discuss the following questions:
  - What are you going to choose?
    - Original Paper on some aspect of Software Testing
    - Project *About* Software Testing
    - Project *Using* Software Testing
  - What is your general idea?
  - What external resources will you need? (e.g., open-source software to contribute to)

**Be prepared to share your answers!**

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## In-Class Activity

Ideas:

- Look into making a tool for desktop UI testing
- Performance testing on existing app
- Unit testing for a Java library
- Unit test creation via LLM

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## Default Project

Create a tests for a sufficiently complicated application *or feature* of your choosing using *each* of the methods covered in class (e.g., boundary value testing, equivalence class testing, code coverage, basis-path testing, ...). Then:

- Compare the utility / benefit of each set of test cases,
- Discuss which tests cases were more *or* less applicable, and
- Identify, with justification / explanation, what set of methods are most applicable.